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United States Patent	12405914
Kind Code	B2
Date of Patent	September 02, 2025
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Display apparatus with power consumption control in relation to external device connections

Abstract

A display apparatus includes a plurality of connectors including a first connector and a second connector, a power supply, a display panel, and circuitry. Each of the plurality of connectors is connectable with one of a plurality of external devices including a first external device and a second external device. The circuitry is to detect that one of the plurality of external devices is connected to one of the plurality of connectors, perform negotiation on electric power with the external device that is detected, and control the power supply to output, based on a result of the negotiation, electric power to be used by the external device that is detected from the connector. In a case that the circuitry detects that the first external device is connected to the first connector and the second external device is connected to the second connector, the circuitry is further to control the power supply to output electric power, input from the first connector connected to the first external device, from the second connector to the second external device. In a case that the electric power to be output from the second connector is determined to be smaller in amount than the electric power to be used by the second external device, the circuitry is to control the power supply to reduce power consumption of the display apparatus to output an amount of the electric power to be used by the second external device from the second connector.

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Appl. No.:	**18/360064**
Filed:	**July 27, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240070108 A1	Feb. 29, 2024

Foreign Application Priority Data

JP	2022-134268	Aug. 25, 2022

Publication Classification

Int. Cl.: **G06F13/42** (20060101); **G06F13/38** (20060101)

U.S. Cl.:

CPC **G06F13/4282** (20130101); **G06F13/382** (20130101); G06F2213/0042 (20130101)

Field of Classification Search

CPC: G06F (13/4282); G06F (13/382); G06F (2213/0042)

USPC: 710/8; 710/14; 710/15; 710/62; 710/104; 710/105; 713/310

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2022-134268, filed on Aug. 25, 2022, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.
BACKGROUND
Technical Field
(2) Embodiments of the present disclosure relate to a display apparatus.
Related Art
(3) In the related art, there is a technology of receiving electric power from an external device or supplying electric power to an external device via an interface such as a universal serial bus (USB).
(4) Using such a technology, a power supply pass-through technology is provided, which supplies electric power from a first external device connected to one interface to a second external device connected to another interface.
SUMMARY
(5) In one aspect, a display apparatus includes a plurality of connectors including a first connector and a second connector, a power supply, a display panel, and circuitry. Each of the plurality of connectors is connectable with one of a plurality of external devices including a first external device and a second external device. The circuitry is to detect that one of the plurality of external devices is connected to one of the plurality of connectors, perform negotiation on electric power with the external device that is detected, and control the power supply to output, based on a result of the negotiation, electric power to be used by the external device that is detected from the connector. In a case that the circuitry detects that the first external device is connected to the first connector and the second external device is connected to the second connector, the circuitry is further to control the power supply to output electric power, input from the first connector connected to the first external device, from the second connector to the second external device. In a case that the electric power to be output from the second connector is determined to be smaller in amount than the electric power to be used by the second external device, the circuitry is to control the power supply to reduce power consumption of the display apparatus to output an amount of the electric power to be used by the second external device from the second connector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A more complete appreciation of embodiments of the present disclosure and many of the

attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:
(2) FIG. **1** is a block diagram illustrating a hardware configuration of a display apparatus according to an embodiment of the present disclosure;
(3) FIG. **2** is a block diagram illustrating a hardware configuration of a video signal conversion cable according to an embodiment of the present disclosure;
(4) FIG. **3** is a block diagram illustrating a functional configuration of a display apparatus according to an embodiment of the present disclosure;
(5) FIG. **4** is a diagram illustrating an overview of power pass-through control performed by a display apparatus according to an embodiment of the present disclosure;
(6) FIG. **5** is a flowchart of the first procedure of processing performed by a display apparatus according to an embodiment of the present disclosure;
(7) FIG. **6** is a diagram illustrating a message displayed by a display apparatus according to an embodiment of the present disclosure;
(8) FIG. **7** is a diagram illustrating a brightness setting screen displayed by a display apparatus according to an embodiment of the present disclosure;
(9) FIG. **8** is a diagram illustrating another brightness setting screen displayed by a display apparatus according to an embodiment of the present disclosure;
(10) FIG. **9** is a diagram illustrating another message displayed by a display apparatus according to an embodiment of the present disclosure;
(11) FIG. **10** is a graph illustrating the relation between the brightness setting value and the power consumption of a display panel included in a display apparatus according to an embodiment of the present disclosure;
(12) FIG. **11** is a flowchart of the second procedure of processing performed by a display apparatus according to an embodiment of the present disclosure; and
(13) FIG. **12** is a flowchart of a third example of processing performed by a display apparatus according to an embodiment of the present disclosure.
(14) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.
DETAILED DESCRIPTION
(15) In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.
(16) Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
(17) Embodiments of the present disclosure are described below with reference to the drawings.
(18) Hardware Configuration of Display Apparatus
(19) FIG. **1** is a block diagram illustrating a hardware configuration of a display apparatus **100** according to the present embodiment.
(20) The display apparatus **100** illustrated in FIG. **1** includes a display panel **102** on a front face of a housing. When the display apparatus **100** is connected to one of external devices **10** (for example, a laptop personal computer (PC) or a smartphone) via one of video signal conversion cables **20** corresponding to the one of the external devices **10** and receives a video signal from the one of the external devices **10** via the one of the video signal conversion cables **20**, the display apparatus **100** displays video on the display panel **102** according to the video signal. The display apparatus **100** is

a thin display apparatus that incorporates a battery **108** and is portable. Accordingly, the display apparatus **100** is carried together with one or more external devices **10**. At the time of use, the display apparatus **100** is connected to one of the one or more external devices **10** to be used as an extended display of the one of the one or more external devices **10**. In the following description, the external devices **10** may be referred to as a single external device **10** unless the external devices **10** are to be distinguished from each other. Similarly, the video signal conversion cables **20** may be referred to as a single video signal conversion cable **20** unless the video signal conversion cables **20** are to be distinguished from each other.
(21) For example, the display apparatus **100** may be used as the extended display by a sales representative when making a presentation at a customer site. Further, for example, the display apparatus **100** may be used as an external display for an external device **10** such as a personal computer (PC) at an office. Furthermore, for example, the display apparatus **100** may be used as an extended display for watching a moving image or working at home when the display apparatus **100** is connected to a smartphone used by an individual outside of an office.
(22) As illustrated in FIG. **1**, the display apparatus **100** includes a controller **101**, the display panel **102**, a communication interface (I/F) **103**, a user interface (UI) **104**, universal serial bus (USB) I/Fs **105**, a power supply **107**, and the battery **108**. In the following description, the USB I/Fs **105** may be referred to as a single USB I/F **105** unless the USB I/Fs **105** are to be distinguished from each other.
(23) The controller **101** controls entire operation of the display apparatus **100**. For example, the controller **101** is implemented by a control circuit such as an integrated circuit (IC) including a central processing unit (CPU), a read only memory (ROM), and a random access memory (RAM).
(24) On the display panel **102**, various images (moving images and still images) are displayed according to video signals transmitted from the controller **101**. A liquid crystal display or an organic electro luminescence (EL) display may be used as the display panel **102**, for example.
(25) The communication I/F **103** is an interface for controlling connection and communication by wireless communication with the external device **10**. The communication I/F **103** is connected to the external device **10** (for example, a laptop PC or a smartphone) by wireless communication, and transmits and receives a control signal and a video signal to and from (performs wireless communication with) the external device **10**. For example, Wireless Fidelity (Wi-Fi) is used as a wireless communication standard used by the communication I/F **103**.
(26) The UI **104** is an interface that receives an input from an operator with an operation performed by the operator. For example, the display apparatus **100** includes a plurality of buttons on side faces and the front face of the housing as the UI **104**. These buttons are, for example, a power button, a select button, an enter button, and a return button. The UI **104** is not limited to these buttons described above. The display apparatus **100** may include another input device (for example, a touch sensor superimposed on the front face of the display panel **102**) as the UI **104**. For example, in response to operations performed by the operator to the UI **104**, the display apparatus **100** turns on and off the power and performs various settings such as settings of brightness and contrast.
(27) The USB I/Fs **105** are interfaces for connecting the external devices **10**. Each of the USB I/Fs **105** includes a USB port **105**A that serves as a “connector” and conforms to the USB standard. A USB cable such as a video signal conversion cable **20** is connected to the USB port **105**A Accordingly, a USB I/F **105** to which the USB port **105**A belongs is connected to the external device **10** via the USB cable. For example, when a communication terminal (for example, a laptop PC or a smartphone) is connected to the USB I/F **105** as the external device **10**, the USB I/F **105** transmits and receives a control signal, a video signal, and electric power to and from the communication terminal. Further, for example, when an external power supply is connected to the USB I/F **105** as the external device **10**, the USB I/F **105** receives electric power supplied from the external power supply.
(28) The display apparatus **100** includes a plurality of USB I/Fs **105**. Multiple external devices **10**

can be separately connected to the plurality of USB I/Fs **105**. In the present embodiment described with reference to FIG. **1**, the display apparatus **100** includes two USB I/Fs **105**. The number of the USB I/Fs **105** is not limited to two. The display apparatus **100** may include three or more USB I/Fs **105**.
(29) The power supply **107** controls the supply of electric power to each component of the display apparatus **100** to drive each component. For example, when electric power is not supplied from the external device **10** (such as an external power supply or a communication terminal) or when electric power supplied from the external device **10** (such as an external power supply or a communication terminal) is insufficient, the power supply **107** can supply electric power stored in the battery **108** to each component of the display apparatus **100**. Further, for example, when electric power is supplied from the external device **10** (such as an external power supply or a communication terminal), the power supply **107** can supply the electric power supplied from the external device **10** to each component of the display apparatus **100**. Furthermore, for example, the power supply **107** can pass through the electric power supplied from the external device **10** (such as an external power supply or a communication terminal) connected to one USB I/F **105** and supply the electric power to another external device **10** (such as a communication terminal) connected to the other USB I/F **105**.
(30) The battery **108** stores electric power to drive the display apparatus **100**. Various rechargeable secondary batteries (for example, a lithium-ion battery and a lithium polymer battery) are usable as the battery **108**. The display apparatus **100** (for example, the power supply **107**) can charge the battery **108** with electric power supplied from the external device **10** (an external power supply or a communication terminal) connected to the USB I/F **105**.
(31) Hardware Configuration of Video Signal Conversion Cable
(32) FIG. **2** is a block diagram illustrating a hardware configuration of the video signal conversion cable **20** according to the present embodiment.
(33) As illustrated in FIG. **2**, the video signal conversion cable **20** includes a HIGH-DEFINITION MULTIMEDIA INTERFACE (HDMI) I/F **21**, a signal converter **22**, a USB I/F **23**, and a power supply **24**.
(34) The HDMI I/F **21** is an interface for connecting the video signal conversion cable **20** to the external device **10** (a communication terminal). For example, the HDMI I/F **21** is connected to an HDMI terminal included in the external device **10**, and transmits and receives a control signal, a video signal, and electric power to and from the external device **10** (a communication terminal).
(35) The signal converter **22** converts the video signal received from the external device **10** via the HDMI I/F **21** into a video signal based on which the display apparatus **100** can display video.
(36) The USB I/F **23** is an interface for connecting the video signal conversion cable **20** to the display apparatus **100**. For example, the USB I/F **23** is connected to one of USB ports **105**A included in the display apparatus **100**, and transmits and receives, for example, a control signal, a video signal, and electric power to and from the display apparatus **100**.
(37) The power supply **24** controls supply of electric power to each component of the video signal conversion cable **20** to drive each component. For example, the power supply **24** supplies electric power supplied from the display apparatus **100** via the USB I/F **23** to each component of the video signal conversion cable **20**.
(38) For example, the video signal conversion cable **20** receives a video signal transmitted from the external device **10** with the HDMI I/F **21** and converts, with the signal converter **22**, the video signal into another video signal based on which the display apparatus **100** can display video. The video signal conversion cable **20** outputs the other video signal converted by the signal converter **22** (i. e., the video signal based on which the display apparatus **100** can display video) to the display apparatus **100** with the USB I/F **23**.
(39) Further, for example, when the display apparatus **100** performs power pass-through control, the video signal conversion cable **20** receives electric power supplied from the display apparatus

100 with the USB I/F **23** and supplies the electric power to each component of the video signal conversion cable **20**.

(40) Functional Configuration of Display Apparatus FIG. **3** is a block diagram illustrating a functional configuration of the display apparatus **100** according to the present embodiment. As illustrated in FIG. **3**, the display apparatus **100** includes a connection detection unit **111**, a negotiation unit **112**, a power output unit **113**, a pass-through control unit **114**, and a display control unit **115**.

(41) The connection detection unit **111** detects that the external device **10** is connected to the USB port **105**A included in the USB I/F **105**. For example, the connection detection unit **111** detects that the external device **10** is connected to the USB port **105**A based on the resistance value of a specific terminal (e.g., a configuration channel (CC) pin) included in the USB port **105**A in accordance with the specification of the USB standard.

(42) The negotiation unit **112** performs negotiation on electric power with the external device **10** connected to the USB port **105**A. For example, the negotiation unit **112** obtains information such as the electric power (voltage and ampere) required by the external device **10** and the electric power (voltage and ampere) that can be output from the external device **10** by performing negotiation with the external device **10**. When the video signal conversion cable **20** is connected to the USB port **105**A, the negotiation unit **112** can perform negotiation with the video signal conversion cable **20**.

(43) The power output unit **113** outputs, based on a result of the negotiation performed by the negotiation unit **112**, the electric power required by the external device **10** connected to the USB port **105**A from the USB port **105**A to which the external device **10** is connected. For example, the power output unit **113** controls the power supply **107** to supply electric power.

(44) When a first external device **10** and a second external device **10** are connected to a first USB port **105**A and a second USB port **105**A, respectively, the pass-through control unit **114** outputs, in accordance with the specification of the USB Power Delivery (USB-PD) standard, the electric power input to the first USB port **105**A through the second USB port **105**A. Accordingly, the pass-through control unit **114** can supply electric power to the second external device **10** connected to the second USB port **105**A to drive the second external device **10**.

(45) At this time, when the electric power that can be output from the second USB port **105**A is determined to be smaller than the electric power required by the second external device **10**, which is specified by the information on the electric power obtained from the second external device **10** by the negotiation unit **112**, the pass-through control unit **114** performs control such that the electric power required by the second external device **10** can be output from the second USB port **105**A by reducing the power consumption of the display apparatus **100**.

(46) Accordingly, even when the electric power input to the first USB port **105**A is relatively small and the electric power that can be output from the second USB port **105**A is determined to be smaller than the electric power required by the second external device **10**, the display apparatus **100** according to the present embodiment can output the electric power required by the second external device **10** from the second USB port **105**A to drive the second external device **10**.

(47) For example, the pass-through control unit **114** performs control such that the electric power required by the video signal conversion cable **20** can be supplied to the video signal conversion cable **20** by lowering an upper limit value of brightness of the display panel **102** included in the display apparatus **100** to reduce the power consumption of the display apparatus **100**.

(48) Accordingly, even when the electric power that can be output from the second USB port **105**A is determined to be smaller than the electric power required by the second external device **10**, the display apparatus **100** according to the present embodiment can output the electric power required by the second external device **10** from the second USB port **105**A by lowering the upper limit value of the brightness of the display panel **102** included in the display apparatus **100** to drive the second external device **10**.

(49) In this case, the pass-through control unit **114** may determine an amount of the upper limit

value of the brightness of the display panel **102** to be lowered according to a shortage amount of the electric power required by the video signal conversion cable **20**. For example, assuming that the shortage amount of the electric power required by the video signal conversion cable **20** is W**1**, the pass-through control unit **114** may determine the amount of the upper limit of the brightness of the display panel **102** to be lowered to be W**1** or W**1**+a. The value α is a certain margin value.
(50) Accordingly, the display apparatus **100** according to the present embodiment can output the electric power required by the second external device **10** from the second USB port **105**A without the upper limit value of the brightness of the display panel **102** being greatly lowered.
(51) Further, for example, the pass-through control unit **114** may perform control such that the electric power required by the video signal conversion cable **20** can be supplied to the video signal conversion cable **20** by turning off a wireless communication function (e.g., the communication I/F **103**) included in the display apparatus **100** to reduce the power consumption of the display apparatus **100**.
(52) Accordingly, even when the electric power that can be output from the second USB port **105**A is determined to be smaller than the electric power required by the second external device **10**, the display apparatus **100** according to the present embodiment can output the electric power required by the second external device **10** from the second USB port **105**A by turning off the wireless communication function included in the display apparatus **100** to drive the second external device **10**.
(53) The USB-PD standard has the following features.
(54) (1) The electric power of 100 watt (W) (20 voltage (V) and 5 ampere (A)) can be supplied at maximum. For this purpose, the output voltage can be varied in a plurality of (four or five) stages.
(55) (2) A device having a plurality of terminals conforming to the USB-PD standard (USB Type-C terminals) receives electric power from one terminal and outputs the electric power from another terminal to supply the electric power to a downstream device in a string manner.
(56) (3) An external device to be connected to the device having the terminals can turn to be both a device that receives electric power and a device that outputs electric power. The electric power can be supplied in both directions.
(57) The display control unit **115** can display various kinds of information on the display panel **102**.
(58) For example, the display control unit **115** displays various setting screens (e.g., a brightness setting screen) on the display panel **102**.
(59) Further, for example, when the display apparatus **100** is switched to a low brightness mode, the display control unit **115** displays, on the display panel **102**, a message for notifying the operator that the display apparatus **100** is to be switched to the low brightness mode.
(60) Furthermore, for example, when the wireless communication function included in the display apparatus **100** is turned off, the display control unit **115** displays, on the display panel **102**, a message for notifying the operator that the wireless communication function included in the display apparatus **100** is to be turned off.
(61) Overview of Power Pass-Through Control Performed by Display Apparatus
(62) FIG. **4** is a diagram illustrating an overview of power pass-through control performed by the display apparatus **100** according to the present embodiment.
(63) In the present embodiment described with reference to each of parts (a) and (b) of FIG. **4**, a smartphone **12** is connected to a first USB port **105**A**1** of the display apparatus **100** via the video signal conversion cable **20**. Accordingly, the display apparatus **100** can display an image on the display panel **102** based on a video signal transmitted from the smartphone **12** via the video signal conversion cable **20**.
(64) Also, in the present embodiment described with reference to each of parts (a) and (b) of FIG. **4**, an alternating current (AC) adapter **11** is connected to a second USB port **105**A**2** of the display apparatus **100**. Accordingly, the display apparatus **100** is driven by the electric power supplied from the AC adapter **11**. The video signal conversion cable **20** can also be driven by a part of the electric

power supplied from the AC adapter **11**, which is to be passed through to the video signal conversion cable **20** by the display apparatus **100**.
(65) The AC adapter **11** according to the present embodiment serves as an external device that serves as a device from which electric power is to be supplied. The video signal conversion cable **20** according to the present embodiment serves as an external device that serves as a device to which electric power is to be supplied.
(66) When the electric power that can be output from the first USB port **105**A**1** is determined to be smaller than the electric power required by the video signal conversion cable **20**, the display apparatus **100** performs control such that the electric power required by the video signal conversion cable **20** can be output from the first USB port **105**A**1** by reducing the power consumption of the display apparatus **100**.
(67) For example, in the present embodiment described with reference to part (a) of FIG. **4**, the electric power supplied from the AC adapter **11** is "15 W" and the power consumption of the display apparatus **100** is "12 W." In this case, the electric power that can be output from the first USB port **105**A**1** is "3 W." Assuming that the electric power required by the video signal conversion cable **20** is "4.5 W," the display apparatus **100** cannot output the electric power "4.5 W" that is required by the video signal conversion cable **20** from the first USB port **105**A**1**. Accordingly, the video signal conversion cable **20** cannot be driven, cannot convert the video signal transmitted from the smartphone **12**, and cannot transmit the converted video signal to the display apparatus **100**.
(68) In this case, the display apparatus **100** reduces the power consumption of the display apparatus **100** as illustrated in part (b) of FIG. **4** by, for example, lowering the upper limit value of the brightness of the display panel **102** included in the display apparatus **100** or by turning off the wireless communication function included in the display apparatus **100**.
(69) For example, in the present embodiment described with reference to part (b) of FIG. **4**, the display apparatus **100** reduces the power consumption of the display apparatus **100** to "10 W." Accordingly, the electric power that can be output from the first USB port **105**A**1** is increased to "5 W." In this way, the display apparatus **100** can output the electric power "4.5 W" required by the video signal conversion cable **20** from the first USB port **105**A**1**. As a result, the video signal conversion cable **20** is driven, converts the video signal transmitted from the smartphone **12**, and transmits the converted video signal to the display apparatus **100**.
(70) First Procedure of Processing Performed by Display Apparatus
(71) FIG. **5** is a flowchart of the first procedure of processing performed by the display apparatus **100** according to the present embodiment.
(72) In the following description, the "USB port **1**" corresponds to the first USB port **105**A (a first connector) included in the display apparatus **100**, and the "USB port **2**" corresponds to the second USB port **105**A (a second connector) included in the display apparatus **100**.
(73) Further, in the following description, it is assumed that the external device **10** connected to the USB port **1** is the second external device **10** (such as the video signal conversion cable **20**) serving as a power supply destination, and it is assumed that another external device **10** connected to the USB port **2** is the first external device **10** (such as the AC adapter **11**) serving as a power supply source.
(74) It is also assumed that the first external device **10** (the AC adapter) is already connected to the USB port **2** before the processing of FIG. **5** starts.
(75) The connection detection unit **111** determines whether the second external device **10** is connected to the USB port **1** (step S**501**). For example, the connection detection unit **111** determines whether the second external device **10** is connected to the USB port **1** based on the resistance value of a specific terminal (e.g., a CC pin) included in the USB port **1** in accordance with the specification of the USB standard.
(76) When it is determined in step S**501** that the second external device **10** is not connected to the

USB port **1** (NO in step S**501**), the connection detection unit **111** executes the processing of step S**501** again.
(77) On the other hand, when it is determined in step S**501** that the second external device **10** is connected to the USB port **1** (YES in step S**501**), the connection detection unit **111** determines whether the second external device **10** connected to the USB port **1** is a sink device (the external device **10** serving as a power supply destination) (step S**502**). For example, the connection detection unit **111** determines whether the second external device **10** connected to the USB port **1** is a sink device based on the resistance value of a specific terminal (e.g., a CC pin) included in the USB port **1** in accordance with the specification of the USB standard. When the video signal conversion cable **20** is connected to the USB port **1** as the second external device **10**, the connection detection unit **111** determines that the second external device **10** connected to the USB port **1** is the sink device.
(78) When it is determined in step S**502** that the second external device **10** connected to the USB port **1** is not a sink device (NO in step S**502**), the negotiation unit **112** performs negotiation with the second external device **10** connected to the USB port **1** in accordance with the specification of the USB-PD standard (step S**510**).
(79) Then, the power output unit **113**, based on a result of the negotiation in step S**510**, outputs electric power to the second external device **10** connected to the USB port **1** or inputs electric power from the second external device **10** connected to the USB port **1** (step S**511**). Subsequently, the display apparatus **100** ends the series of the processing illustrated in FIG. **5**.
(80) On the other hand, when it is determined in step S**502** that the second external device **10** connected to the USB port **1** is a sink device (YES in step S**502**), the connection detection unit **111** determines whether the first external device **10** is connected to the USB port **2** (step S**503**). For example, the connection detection unit **111** determines whether the first external device **10** is connected to the USB port **2** based on the resistance value of a specific terminal (e.g., a CC pin) included in the USB port **2** in accordance with the specification of the USB standard.
(81) When it is determined in step S**503** that the first external device **10** is not connected to the USB port **2** (NO in step S**503**), the display apparatus **100** ends the series of the processing illustrated in FIG. **5**.
(82) On the other hand, when it is determined in step S**503** that the first external device **10** is connected to the USB port **2** (YES in step S**503**), the negotiation unit **112** performs negotiation with the first external device **10** connected to the USB port **2** in accordance with the specification of the USB-PD standard to determine whether the first external device **10** connected to the USB port **2** can output electric power of 5 V (i. e., the voltage required by the second external device **10** connected to the USB port **1**) (step S**504**).
(83) At this point, the negotiation unit **112** does not perform negotiation with the second external device **10** connected to the USB port **1**, but uses electric power of a particular voltage value that is the electric power of 5 V required by the second external device **10** connected to the USB port **1** to drive the second external device **10** connected to the USB port **1**. The particular value is stored in a memory in advance.
(84) When it is determined in step S**504** that the first external device **10** connected to the USB port **2** cannot output the electric power of 5 V (NO in step S**504**), the display apparatus **100** ends the series of the processing illustrated in FIG. **5**.
(85) On the other hand, when it is determined in step S**504** that the first external device **10** connected to the USB port **2** can output the electric power of 5 V (YES in step S**504**), the pass-through control unit **114** determines whether the power consumption of the display apparatus **100** is to be reduced (step S**505**).
(86) In step S**505**, the pass-through control unit **114** determines that the power consumption of the display apparatus **100** is to be reduced when the following formula (1) is satisfied.
(87) Electric power supplied from the first external device **10** connected to the USB port **2**

Power consumption of the display apparatus 100<Electric power required by the second external device 10 connected to the USB port 1 Formula (1):
(88) In other words, the pass-through control unit **114** reduces the power consumption of the display apparatus **100** on the condition that the value of the electric power obtained by subtracting the power consumption of the display apparatus **100** from the value of the electric power supplied from the first external device **10** connected to the USB port **2** is smaller than the value of the electric power required by the second external device **10** connected to the USB port **1**.
(89) When it is determined in step S**505** that the power consumption of the display apparatus **100** is to be reduced (YES in step S**505**), the pass-through control unit **114** switches the display apparatus **100** to a power consumption reduction mode to reduce the power consumption of the display apparatus **100** (step S**506**). Subsequently, the display apparatus **100** proceeds the processing to step S**507**.
(90) On the other hand, when it is determined in step S**505** that the power consumption of the display apparatus **100** is not to be reduced (NO in step S**505**), the display apparatus **100** proceeds the processing to step S**507**.
(91) In step S**507**, the negotiation unit **112** performs negotiation with the first external device **10** connected to the USB port **2** to change the electric power output from the first external device **10** connected to the USB port **2** to the electric power of 5 V (step S**507**).
(92) Specifically, the negotiation unit **112** notifies the sink device (the second external device **10** connected to the USB port **1**) of a plurality of voltage values that the source device (the first external device **10** connected to the USB port **2**) can output. The negotiation unit **112** receives, from the sink device, a notification of a voltage value, i.e., 5 V in the present embodiment, required by the sink device among the voltage values. Further, the negotiation unit **112** notifies the source device to output the voltage value, i.e., 5 V in the present embodiment, required by the sink device. In this way, the source device starts supplying electric power having the voltage value required by the sink device.
(93) The pass-through control unit **114** turns on the function of the power pass-through from the USB port **2** to the USB port **1** (step S**508**).
(94) As a result, a part of the electric power of 5 V input from the USB port **2** is output from the USB port **1** as the electric power required by the second external device **10** connected to the USB port **1**, and the power of the second external device **10** connected to the USB port **1** is turned on (step S**509**).
(95) For example, it is assumed that the second external device **10** connected to the USB port **1** is the video signal conversion cable **20**. When the power of the video signal conversion cable **20** is turned on, the video signal conversion cable **20** can start conversion of a video signal transmitted from a communication terminal such as a smartphone and transmission of the converted video signal to the display apparatus **100**.
(96) Subsequently, the display apparatus **100** ends the series of the processing illustrated in FIG. **5**.
(97) Message and Brightness Setting Screen Displayed by Display Apparatus
(98) FIGS. **6** and **9** are diagrams each illustrating a message displayed by the display apparatus **100** according to the present embodiment. FIGS. **7** and **8** are diagrams each illustrating a brightness setting screen displayed by the display apparatus **100** according to the present embodiment.
(99) As described above, when the electric power that can be output from the first USB port **105**A is determined to be smaller than the electric power required by the second external device **10**, the display apparatus **100** according to the embodiment is switched to the low brightness mode from a normal mode. Accordingly, the display apparatus **100** according to the embodiment can reduce the power consumption of the display apparatus **100** to output the electric power required by the second external device **10** from the first USB port **105**A.
(100) In this case, as illustrated in FIG. **6**, the display control unit **115** displays, on the display panel **102**, a message **601** for notifying the operator that the display apparatus **100** is to be switched to the

low brightness mode.
(101) As illustrated in FIGS. **7** and **8**, in response to an operation to the UI **104** performed by the operator, the display control unit **115** displays, on the display panel **102**, a brightness setting screen **701** that indicates a setting value of the brightness of the display panel **102** in a bar graph.
(102) In FIG. **7**, the brightness setting screen **701** when the display apparatus **100** is in the normal mode is illustrated. In FIG. **8**, the brightness setting screen **701** when the display apparatus **100** is in the low brightness mode is illustrated. The brightness setting screen **701** illustrated in FIG. **8** presents that the upper limit value of the brightness of the display panel **102** is lowered.
(103) As illustrated in FIG. **8**, when the upper limit value of the brightness of the display panel **102** is lowered and the operator performs an operation to increase the brightness of the display panel **102** beyond the upper limit value, the pass-through control unit **114** invalidates the operation.
(104) In an alternative embodiment of the present disclosure, when the electric power that can be output from the first USB port **105**A is determined to be smaller than the electric power required by the second external device **10**, the display apparatus **100** according to the present embodiment turns off the wireless communication function (e.g., the communication I/F **103**) included in the display apparatus **100**. Accordingly, the display apparatus **100** according to the embodiment can reduce the power consumption of the display apparatus **100** to output the electric power required by the second external device **10** from the first USB port **105**A.
(105) In this case, as illustrated in FIG. **9**, the display control unit **115** displays, on the display panel **102**, a message **602** for notifying the operator that the wireless communication function included in the display apparatus **100** is to be turned off.
(106) Relation Between Brightness Setting Value and Power Consumption of Display Panel
(107) FIG. **10** is a graph illustrating the relation between the brightness setting value and the power consumption of the display panel **102** included in the display apparatus **100** according to the present embodiment.
(108) In the graph illustrated in FIG. **10**, the vertical axis represents power consumption [W] of the display panel **102** and the horizontal axis represents the brightness setting value of the display panel **102**.
(109) As illustrated in FIG. **10**, the brightness setting value of the display panel **102** can be set in a range from “0” to “100.”
(110) The power consumption of the display panel **102** increases in proportion to the brightness setting value. For this reason, the display apparatus **100** according to the embodiment can reduce the power consumption of the display apparatus **100** by lowering the upper limit value of the brightness setting value of the display panel **102**.
(111) For example, in the present embodiment with reference to FIG. **10**, the power consumption of the display panel **102** is obtained by the following formula (2).
Power consumption $[W]=\frac{1}{2}\times$Brightness setting value$+5$ Formula (2):
Second Procedure of Processing Performed by Display Apparatus
(112) FIG. **11** is a flowchart of the second procedure of processing performed by the display apparatus **100** according to the present embodiment.
(113) The flowchart of FIG. **11** differs from the flowchart of FIG. **5** in that step S**521** is included between step S**504** and step S**505**.
(114) In step S**521**, the pass-through control unit **114** determines whether a remaining amount of the battery **108** included in the display apparatus **100** is smaller than a certain threshold value (step S**521**). As the certain threshold value, at least a value of the remaining amount sufficient to drive the display apparatus **100** with the battery is used.
(115) When it is determined in step S**521** that the remaining amount of the battery is smaller than the certain threshold value (YES in S**521**), the display apparatus **100** proceeds the processing to step S**505**.
(116) On the other hand, when it is determined in step S**521** that the remaining amount of the

battery is not smaller than the certain threshold value (NO in step S**521**), the display apparatus **100** proceeds the processing to step S**507**.
(117) It is assumed that the remaining amount of the battery **108** is sufficient to drive the display apparatus **100** with the battery (the remaining amount is equal to or greater than the certain threshold value). According to the flowchart of FIG. **11**, even when the electric power that can be output from the USB port **1** is determined to be smaller than the electric power required by the second external device **10** connected to the USB port **1**, the display apparatus **100** can output the electric power required by the second external device **10** connected to the USB port **1** from the USB port **1** without consuming the electric power input from the USB port **2** by being driven with the battery **108**. In this way, the electric power required by the second external device **10** connected to the USB port **1** can be output from the USB port **1**. As a result, according to the flowchart of FIG. **11**, the frequency of lowering the power consumption of the display apparatus **100** is reduced, and thus a decrease of operability for the operator is prevented or eliminated.
(118) When the remaining amount of the battery **108** falls below the certain threshold value while the display apparatus **100** is being driven with the battery, the display apparatus **100** may stop driving the display apparatus **100** with the battery and switch the display apparatus **100** to the power consumption reduction mode to reduce the power consumption of the display apparatus **100**. Accordingly, the display apparatus **100** can continuously supply the electric power to the second external device **10** connected to the USB port **1**.
(119) Third Procedure of Processing Performed by Display Apparatus
(120) FIG. **12** is a flowchart of the third procedure of processing performed by the display apparatus **100** according to the present embodiment.
(121) The flowchart of FIG. **12** differs from the flowchart of FIG. **5** in that steps S**531** to S**535** are included after step S**509**.
(122) In step S**531**, the pass-through control unit **114** determines whether the second external device **10** connected to the USB port **1** has a video output function (step S**531**).
(123) When it is determined in step S**531** that the second external device **10** connected to the USB port **1** has the video output function (YES in step S**531**), the display apparatus **100** ends the series of the processing illustrated in FIG. **12**.
(124) On the other hand, when it is determined in step S**531** that the second external device **10** connected to the USB port **1** does not have the video output function (NO in step S**531**), the pass-through control unit **114** determines whether the display apparatus **100** is in the power consumption reduction mode (step S**532**).
(125) When it is determined in step S**532** that the display apparatus **100** is not in the power consumption reduction mode (NO in step S**532**), the display apparatus **100** ends the series of the processing illustrated in FIG. **12**.
(126) On the other hand, when it is determined in step S**532** that the display apparatus **100** is in the power consumption reduction mode (YES in step S**532**), the pass-through control unit **114** turns off the function of the power pass-through from the USB port **2** to the USB port **1** (step S**533**).
(127) The negotiation unit **112** performs negotiation with the first external device **10** connected to the USB port **2** to change the electric power output from the first external device **10** connected to the USB port **2** to the voltage before being changed to 5 V (step S**534**).
(128) The pass-through control unit **114** switches the display apparatus **100** from the power consumption reduction mode to the normal mode (step S**535**). Subsequently, the display apparatus **100** ends the series of the processing illustrated in FIG. **12**.
(129) It is assumed that the second external device **10** connected to the USB port **1** does not have the video output function. According to the flowchart of FIG. **12**, even when the electric power that can be output from the USB port **1** is determined to be smaller than the electric power required by the second external device **10** connected to the USB port **1**, the display apparatus **100** is not switched to the power consumption reduction mode and does not output the electric power from the

USB port **1**. As a result, according to the flowchart of FIG. **12**, the frequency of lowering the power consumption of the display apparatus **100** is reduced, and thus a decrease of operability for the operator is prevented or eliminated.

(130) According to the above-described embodiment, screen display of the display apparatus **100** using a communication terminal and a video signal conversion cable can be performed in various environments. There is a case where the operator desires to use the display apparatus **100** in a scene where the environment changes from time to time. Such a scene where the environment changes from time to time includes, for example, a scene in which the location of the operator holding the display apparatus **100** is changed, a scene in which a communication terminal (such as a laptop PC, a smartphone, or a tablet PC) that presents a screen to be displayed on the display apparatus **100** is changed, and a scene in which a video signal conversion cable or an AC adapter to be connected to the display apparatus **100** for use is changed. The display apparatus **100** can display a screen presented by the communication terminal in many scenes even in such a case.

(131) The above-described embodiment is illustrative and does not limit the present disclosure. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present disclosure.

(132) For example, the “connector” is not limited to a “USB” port, and may be any connector that allows at least a function of power pass-through.

(133) Further, for example, the method of “reducing the power consumption of the display apparatus” is not limited to the method of lowering the upper limit value of the brightness of the display panel or the method of turning off the wireless communication function included in the display apparatus. Any method may be used as long as the method reduces at least the power consumption of the display apparatus.

(134) The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, application specific integrated circuits (ASICs), digital signal processors (DSPs), field programmable gate arrays (FPGAs), conventional circuitry and/or combinations thereof which are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carries out or is programmed to perform the recited functionality. The hardware may be any hardware disclosed herein or otherwise known which is programmed or configured to carry out the recited functionality. When the hardware is a processor which may be considered a type of circuitry, the circuitry, means, or units are a combination of hardware and software, the software being used to configure the hardware and/or processor.

Claims

1. A display apparatus comprising: a display panel configured to display an image; a plurality of connectors including a first connector and a second connector, the plurality of connectors each being connectable with one of a plurality of external devices including a first external device and a second external device, the first external device being a device from which electric power is to be supplied, and the second external device being a device to which electric power is to be supplied; and circuitry configured to: detect that one of the plurality of external devices is connected to one of the plurality of connectors; perform negotiation on electric power supplied from the first external device and required for the second external device to operate; and control power consumption of the display apparatus such that part of the electric power supplied from the first external device to the display apparatus is passed to the second external device, based on a result of the negotiation, wherein, in a case in which the circuitry detects that the first external device is connected to the

first connector and the second external device is connected to the second connector, the circuitry is further configured to: determine whether the power consumption of the display apparatus needs to be reduced based on the electric power supplied from the first external device and required for the second external device; and in a case in which the circuitry determines that the power consumption of the display apparatus needs to be reduced, reduce the power consumption of the display apparatus by an amount of the electric power to satisfy a requirement for the second external device to operate, and pass the amount of the electric power to the second external device.
2. The display apparatus according to claim 1, wherein the circuitry is configured to lower an upper limit value of brightness of the display panel to reduce the power consumption of the display apparatus.
3. The display apparatus according to claim 2, wherein the circuitry is configured to determine an amount of the upper limit value of the brightness of the display panel to be lowered in accordance with a shortage amount of the electric power to be used by the first external device.
4. The display apparatus according to claim 2, wherein the circuitry is further configured to, in a case in which the upper limit value of the brightness of the display panel is lowered, invalidate an operation for increasing the brightness of the display panel beyond the upper limit value.
5. The display apparatus according to claim 1, further comprising: a communication interface, wherein the circuitry is configured to turn off a wireless communication function of the communication interface, so as to reduce the power consumption of the display apparatus.
6. The display apparatus according to claim 1, further comprising a battery to supply electric power to drive the display apparatus, wherein the circuitry is further configured to determine whether a remaining amount of the battery is equal to or greater than a threshold value, in a case in which the electric power to be output from the second connector is determined to be smaller than the electric power to be used by the second external device, and based on a determination that the remaining amount of the battery is equal to or greater than the threshold value, drive the display apparatus with the battery, and output, from the second connector, the electric power input from the first connector without reducing the power consumption of the display apparatus.
7. The display apparatus according to claim 1, wherein the circuitry is further configured to determine whether the second external device has a video output function, and based on a determination that the second external device does not have the video output function, prevent output of electric power from the second connector, without reducing the power consumption of the display apparatus.
8. The display apparatus according to claim 1, wherein the circuitry is further configured to, in a case in which connection of the second connector with the second external device is detected after connection of the first connector with the first external device is detected, perform negotiation with the first external device at a time when the connection of the second connector with the second external device is detected, and cause the first external device to output electric power having a voltage value compatible with the second external device.
9. The display apparatus according to claim 1, wherein the circuitry is further configured to, in a case in which the second external device that is connected is determined to be a power supply destination, output electric power of a particular voltage value sufficient to drive the second external device from the second connector without performing negotiation with the second external device.
10. The display apparatus according to claim 1, wherein the plurality of connectors are capable of transmitting or receiving data to and from the plurality of external devices connected to the plurality of connectors, in addition to input or output of electric power.
11. The display apparatus according to claim 1, wherein at least one of the plurality of connectors is a universal serial bus (USB) port.
12. The display apparatus according to claim 1, wherein the circuitry is further configured to

display, on the display panel, a message for notifying an operator that the power consumption of the display apparatus is to be reduced.